

Projects, Big and Small

. Small Artist Website

Tech Stack: HTML/CSS, JavaScript (or CMS/Static Site Tool)

- **Overview:** A tailored web presence for a small artist to display media and portfolio items.
- **Key Components:** Visual layout, media hosting, responsive UI, and client specification mapping.

2. Class Diagrams & System Design

Language / Notation: UML

- **Overview:** Architectural planning and object-oriented structural modeling for application design (e.g., medicine dispensing system).
- **Key Components:** Class diagrams, inheritance, relationships, and system domain modeling prior to code execution.

3. Unix To-Do List CLI

Environment: Ubuntu Shell (Bash Scripting / Command Line)

- **Overview:** A command-line task management tool tailored for Unix environments.
- **Key Components:** Shell scripting, file manipulation, and system documentation formatted with standard Unix `man` page styling.

4. Java Budgeting System

Language: Java

- **Overview:** A file-handling application designed for tracking personal or organizational expenses.
- **Key Components:**
 - **Core Features:** Expense loading/saving, annual/monthly spend calculations, average spend per item, and peak spending month identification.
 - **Advanced Processing:** File I/O paired with encryption/decryption pipelines for secure storage.

5. Java Scrabble Game

Language: Java

- **Overview:** A full digital implementation of the classic Scrabble board game.
- **Key Components:** Board state management, tile collection, turn execution logic, score calculation, and word validation rules.

6. Java Stock Management Binary Tree

Language: Java

- **Overview:** A data-structure-driven application for inventory and asset tracking using binary search trees.
- **Key Components:**
 - **Tree Operations:** Insertion, search, node deletion, and level-order traversal.
 - **Financial Metrics:** Dynamic total asset value evaluation across the node network.

7. MySQL Relational Sports Club Database

Tech Stack: MySQL / Relational SQL

- **Overview:** A relational database system designed for multi-facility racquet sports institutions (badminton and tennis) managing staff, facilities, and maintenance.
- **Key Components:**
 - **Comprehensive Schema:** Tables spanning `player`, `coach`, `facility_manager`, `stringer`, `facility`, `court`, `sessions`, `invoice`, `booking`, `racquet`, and `stringing_job`.
 - **Relationships & Execution:** Dynamic table relationships using entity join tables (`coach_player`, `coach_facility`, `facility_manager_facility`, `stringer_facility`), column alterations, and constraint setup scripts.

8. MongoDB NoSQL Coach Booking Platform

Tech Stack: MongoDB, JSON, Aggregation Pipelines

- **Overview:** A NoSQL document-based database platform built for sports coaching bookings, user management, and rating aggregations.
- **Key Components:**
 - **Schema Design:** Multi-collection platform (`users`, `sports`, `coachAvailability`, `bookings`) utilizing embedded profiles and time-slots alongside referenced foreign object keys (`$oid`).
 - **Role-Based Modeling:** Role encapsulation (`athlete`, `coach`, `admin`) supporting custom access queries, upcoming/completed booking views, and availability lookups.
 - **Advanced Analytics & Performance:** Aggregation pipelines for calculated metrics (coach revenue, session volume per sport, and dynamically updated average ratings) backed by multi-field database indexing.

9. C++ Library Management System

Language: C++

- **Overview:** An object-oriented software system for managing library resources and operations.

- **Key Components:** Multiple interacting classes leveraging inheritance, encapsulation, polymorphism, dynamic memory allocation, and data persistence.

10. Interactive Team Web Application

Tech Stack: Flutter, React, Web Frameworks

- **Overview:** A multi-framework interactive website built collaboratively in a cross-functional team environment.
- **Key Components:** Cross-platform React and Flutter UI integration, dynamic user state management, and team-based version control workflows.

11. C Dijkstra Graph & Shortest Path Solver

Language: C

- **Overview:** A low-level system implementing graph theory and pathfinding algorithms with manual memory management.
- **Key Components:**
 - **Graph Structures:** Conversions between adjacency matrices and adjacency lists.
 - **Operations & Traversal:** Single/batch edge insertion, file-based loading/saving, Depth-First Search (DFS) traversal.
 - **Pathfinding:** Shortest path computation using Dijkstra's algorithm.

12. AI Warehouse Grid Pathfinding Agent

Tech Stack: Python, PyTorch

- **Overview:** An artificial intelligence model engineered to navigate grid environments and compute optimal navigation paths for automated warehouse logistics.
- **Key Components:** Deep learning / PyTorch framework implementations, spatial constraint navigation, grid-based optimization, and decision logic.

In Progress: Next-Gen Badminton Mobile/Web App

- **Status:** Active Development (*Projected to be the largest, most complex full-stack project*)
- **Overview:** A feature-rich ecosystem combining complex domain logic, racquet sports management, real-time sync, and scalable mobile/web platforms.